Capacity Allocation Decision

Workload **wl-2026-0842** · facility iad-dc-01, Ashburn VA · conditions 'normal' · trace gm-05662fb1cfed

APPROVED WITH CONDITIONS	Zone **zone-c**	Delay **14.0 h**	Rounds **3**	Capex **$48,000**

Plan

Deploy the 6 GB200 NVL72 racks in Zone-C. Utilize 5 existing liquid-ready racks and retrofit 1 additional rack to meet the 6-rack requirement, incurring a 14-hour retrofit delay and a one-time capex of $48,000.00.

Conditions

- Retrofit 1 rack in Zone-C for direct-to-chip liquid cooling
- Accept a 14-hour installation delay for the retrofit
- Stagger installation shifts using the 5 liquid-certified technicians on shift (2 per rack, max 4 concurrent installs)

How the conflicting verdicts were reconciled

The conflict was resolved by identifying Zone-C as the only physically viable zone. Zone-A is ruled out due to being air-cooled and having a floor load limit of 1200.0 kg/rack (below the required 1360.0 kg/rack). Zone-B is ruled out because it has only 560.6 kW of power headroom (792.0 kW required) and only 2 free CDU ports (6 required). The Cost agent preferred Zone-B to avoid capex, but this is physically impossible. The Facilities agent unlocked the solution by revealing that Zone-C only requires retrofitting 1 rack (not all 6), which reduces the capex from the Cost agent's feared $288,000.00 to just $48,000.00, with a manageable 14-hour delay.

Trade-offs accepted

- Gained physical feasibility in Zone-C (3053.0 kW power headroom, 10 CDU ports, and 1800.0 kg/rack floor limit) vs Zone-B which lacks power (560.6 kW) and CDU ports (2).
- Incurred a 14-hour delay and $48,000.00 capex to retrofit 1 rack in Zone-C, instead of zero retrofit cost in Zone-B.
- Avoided placing in Zone-A which is air-cooled and has a legacy floor load limit of 1200.0 kg/rack (workload requires 1360.0 kg/rack and liquid cooling).

Cost and efficiency

Measure	Value	Reported by
Recurring monthly cost	**not reported**	cost agent
Change vs original request	**-5.0%**	cost agent
One-time capex	**$48,000**	cost agent
Monthly saving vs next-best	**not reported**	cost agent
Payback period	**not reported**	cost agent
PUE at chosen zone	**1.226**	cooling agent
PUE at next-best zone	—	cooling agent

Zone-C provides highly efficient direct-to-chip liquid cooling with an estimated PUE of 1.226, well below the 1.60 maximum threshold.

Stranded capacity avoided

Avoided stranding 560.6 kW of power headroom in Zone-B which cannot be cooled due to having only 2 free CDU ports, and avoided stranding power in Zone-A which cannot support the 1360.0 kg rack weight or liquid cooling.

The negotiation, round by round

Round 1 is independent: no agent sees another's answer. Later rounds re-prompt each agent with the others' positions attached.

Round 1 — joint feasibility: **hard_refusal**

Agent	Verdict	Zone	Reasoning
cooling	**infeasible**	—	The cooling agent could not produce a trustworthy verdict after 3 attempts (RuntimeError). Failing safe to infeasible and escalating for human review rather than guessing.
cost	**infeasible**	—	The cost agent could not produce a trustworthy verdict after 3 attempts (RuntimeError). Failing safe to infeasible and escalating for human review rather than guessing.
facilities	**infeasible**	—	The facilities agent could not produce a trustworthy verdict after 3 attempts (RuntimeError). Failing safe to infeasible and escalating for human review rather than guessing.
power	**feasible**	zone-a	Zone-A has 3887.2 kW of electrical headroom and 388 spare 30A/208V circuits, easily supporting the 792.0 kW request (6 racks at 132.0 kW each, requiring roughly 80 circuits). Substation load of 42.47 MW plus the added load remains well below 85% of the 72.0 MW firm capacity. No planned outages affect Zone-A's upstream switchgear SG-1.

Round 2 — joint feasibility: **zone_mismatch**

Agent	Verdict	Zone	Reasoning
cooling	**conditional**	zone-c	The power agent proposes zone-a, but zone-a is an air-cooled CRAH zone with a hard ceiling of 40.0 kW/rack and zero free CDU ports. The requested workload is direct_to_chip_liquid requiring 132.0 kW per rack across 6 racks (total heat load approx 776.2 kW, which is 0.98 x 792.0 kW). Zone-c is a liquid DLC zone with a max of 140.0 kW/rack, 10 free CDU ports, and 4129.9 kW of thermal headroom, making it fully capable of supporting the per-rack density and total heat load with an estimated PUE of 1.226. Therefore, the request is conditional upon moving placement from zone-a to zone-c.
cost	**feasible**	zone-a	Zone-A is financially feasible for the 792 kW request. The monthly operating cost for Zone-A is efficient with a cost of 12.4 USD per kW-month, and total facility budget remaining is 2262244.10 USD against a monthly opex budget of 7200000.00 USD. Current cost per GPU-hour is 2.078 USD, which remains below the target of 2.20 USD and maximum of 2.90 USD. Virginia sales-and-use tax exemption (Va. Code Ann. Sec. 58.1-609.3(18)) lowers effective capex on qualifying equipment.
facilities	**conditional**	zone-b	Workload wl-2026-0842 requires 6 racks of GB200 NVL72, with each rack weighing 1,360 kg and requiring direct-to-chip liquid cooling. Zone-B has 8 liquid-ready racks free, and its floor load limit is 1,800 kg per rack (exceeding the 1,360 kg requirement). Zone-B can fully host the 6 racks immediately with available liquid-ready racks. However, power agent endorsed Zone-A, but Zone-A has a legacy floor load limit of 1,200 kg per rack (failing the 1,360 kg workload weight) and 0 liquid-ready racks. Therefore, Zone-B is the physically viable zone for this liquid-cooled high-weight workload. Crew c
power	**feasible**	zone-a	Zone A has 3887.2 kW of electrical headroom, 388 spare 30A/208V circuits, and no planned outages, easily supporting the 792.0 kW request (6 racks at 132.0 kW each, requiring roughly 80 circuits). Substation load of 42.47 MW plus the added load remains well within the 72.0 MW firm capacity limit.

Round 3 — joint feasibility: **zone_mismatch**

Agent	Verdict	Zone	Reasoning
cooling	conditional	zone-c	The workload requires direct-to-chip liquid cooling for 6 racks at 132.0 kW per rack (total thermal load of ~776.2 kW). Power and Facilities agents suggested Zone-A and Zone-B, but Zone-A is an air-cooled CRAH zone with a hard ceiling of 40.0 kW per rack, making it thermally impossible. Zone-B is a liquid DLC zone with a 140.0 kW per rack limit, which can handle the per-rack density, but it only has 2 free CDU ports while 6 are required. Zone-C is the optimal choice as a liquid DLC zone with a 140.0 kW per rack limit, 4,129.9 kW of thermal headroom, and 10 free CDU ports available (exceeding t
cost	conditional	zone-b	Placing the 6 liquid-cooled GB200 racks requires direct-to-chip cooling, ruling out zone-a (air-cooled). Between the feasible liquid-ready options proposed by cooling (zone-c) and facilities (zone-b), zone-b is the superior financial choice because zone-b already has existing DLC infrastructure with zero liquid retrofit cost per rack ($0), whereas zone-c incurs a heavy retrofit cost of $48,000.00 per rack ($288,000.00 total for 6 racks). Under Virginia's sales-and-use tax exemption (Va. Code Ann. Sec. 58.1-609.3(18)), qualifying equipment purchases avoid sales tax, but avoiding the capex entir
facilities	conditional	zone-c	Workload wl-2026-0842 requires 6 racks of GB200 NVL72 at 1360.0 kg each with direct_to_chip_liquid cooling. Cooling agent proposes zone-c. Zone-C has 8 available racks, 5 liquid-ready and 3 retrofittable. Floor load limit is 1800.0 kg/rack, which easily supports the 1360.0 kg rack weight. To meet the 6-rack requirement, 1 rack must be retrofitted for DLC taking 14.0 hours. Certified technicians on shift: 7, liquid-certified: 5, crew required: 2 per rack, max concurrent installs: 4.
power	feasible	zone-c	Zone-c has 3053.0 kW of electrical headroom and 305 spare 30A/208V circuits, which easily accommodates the 792.0 kW request (requiring approximately 80 circuits at 10 kW per circuit). Zone-b, while suggested by facilities, only has 560.6 kW of headroom remaining, which is insufficient for the 792.0 kW load.

Provenance

The orchestrator holds no access to the power, cooling, facilities or cost databases — enforced by IAM Conditions, not convention. Every figure in this report was reported by a specialist agent and is attributable to its verdict above.

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gcloud logging read 'jsonPayload.correlation_id="gm-05662fb1cfed"'
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